

List 1 - Question 11

Kainã Alves Tureso - 15466391

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Question 11

Let v be a test function such that $v(0) = 0$. We have:

$$-\frac{d}{dx} \left(\mu \frac{du}{dx} \right) v dx + \zeta u = F \implies \left(-\frac{d}{dx} \left(\mu \frac{du}{dx} \right) + \zeta u \right) v = Fv \quad (1)$$

Integrating both sides:

$$-\int_0^L \frac{d}{dx} \left(\mu \frac{du}{dx} \right) v dx + \zeta \int_0^L u v dx = \int_0^L F v dx$$

Integrating by parts the first integral gives us:

$$-\frac{d}{dx} \left(\mu \frac{du}{dx} \right) v dx = -\mu v \frac{du}{dx} \Big|_0^L + \int_0^L \mu \frac{du}{dx} \frac{dv}{dx} dx$$

As $v(0) = 0$:

$$-\mu v \frac{du}{dx} \Big|_0^L = -\mu \frac{du}{dx}(L) v(L) = H_L v(L)$$

Substituting back, gives us the variational form:

$$\int_0^L \left(\mu \frac{du}{dx} \frac{dv}{dx} + \zeta uv \right) dx = \int_0^L F v dx - H_L v(L) \quad (2)$$

(i) The geometric simplifications: We are assuming the pipe is very long (the flow is fully developed), has a constant diameter and lies in $\mathbb{R}^2$ (We have only one degree of freedom across the diameter).

(ii) Significance of boundary conditions:

- $u(0) = G_0$: we're setting the velocity at boundary. If $G_0 = 0$, then it's a no-slip condition.
- $-\mu \frac{du}{dx}(L) = H_L$: According to the book "Introduction to Fluid Dynamics" by Robert W. Fox, $\mu \frac{du}{dx}$ is the shear stress. So, if $H_L \neq 0$, it means the pipe is moving at $x = L$

(iii) I don't really know. Google's Gemini answered:

"Physical Origin of the Reaction Term (ζu): Standard Navier-Stokes equations for channel flow only contain the viscous term and a pressure gradient source term (F). The linear reaction term ζu represents a body force proportional and opposite to the velocity. Physically, this represents Darcy friction/drag, which occurs if the pipe or channel is filled with a porous medium that impedes the fluid flow."